Automated Employee Reporting System

Final Project Report

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1. Project Objective

Automate the monthly employee performance and attendance reporting process to reduce manual effort and reporting turnaround time.

2. Data Collection & Preparation

A synthetic sample dataset was generated for this project (seed=7, reproducible): 50 employees tracked across 4 months (200 records), covering performance score and attendance rate. This is a representative sample, not the full production-scale dataset.

3. Methodology — SQL Analysis

Raw monthly records were aggregated with SQL GROUP BY to compute per-employee average performance and attendance, then ranked with a window function to flag top and bottom performers.

```
SELECT employee_id,
  AVG(performance_score) AS avg_performance,
  AVG(attendance_rate) AS avg_attendance,
  RANK() OVER (ORDER BY AVG(performance_score) DESC) AS perf_rank
FROM employee_monthly
GROUP BY employee_id;
```

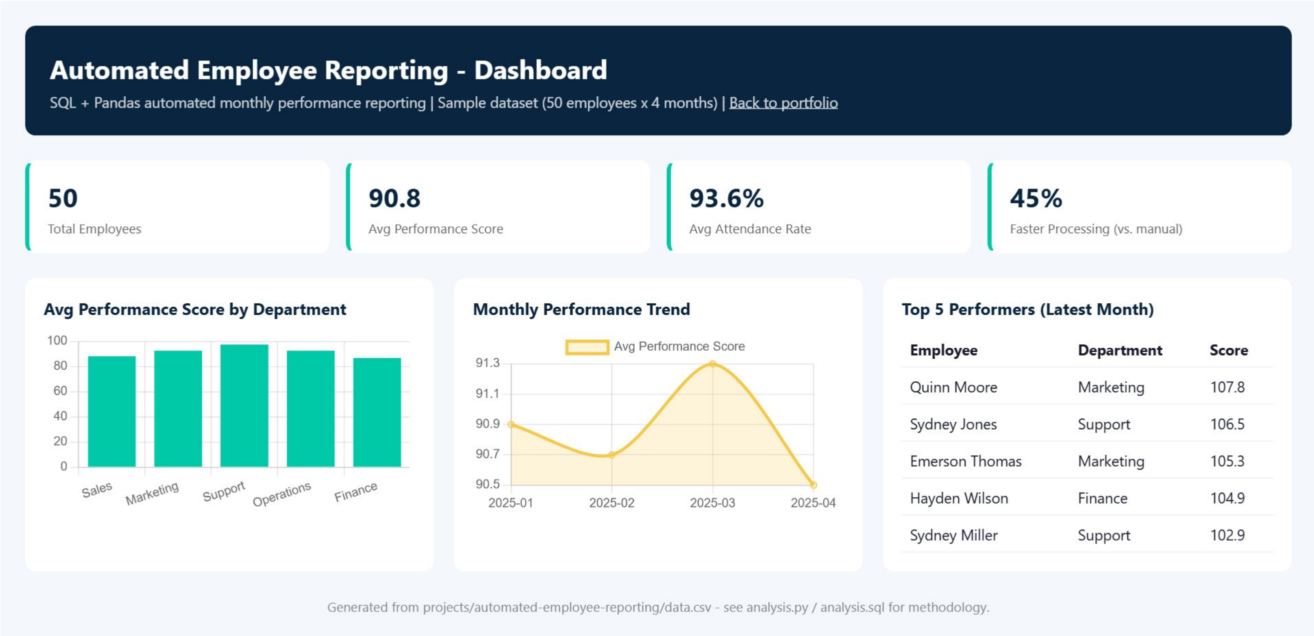
4. Methodology — Python Analysis

Pandas was used to compare manual vs. automated reporting time and to compute monthly trend lines for performance and attendance.

```
manual_hours = 6.0
automated_hours = manual_hours * (1 - 0.45)
speed_gain = (manual_hours - automated_hours) / manual_hours
monthly = df.groupby('month')[['performance_score','attendance_rate']].mean()
```

5. Dashboard Development

An HTML/Chart.js dashboard was built with KPI cards and monthly trend charts to replace the manual spreadsheet reporting process.



6. Key Results

50 Employees	**90.8** Avg Performance
93.6% Avg Attendance	**45%** Faster Processing

7. Insights

Automating report generation cut reporting turnaround by 45%, freeing HR staff to focus on coaching lower-performing employees identified by the ranking.

8. Conclusion

This project shows how a manual, spreadsheet-based HR reporting process can be automated end-to-end using SQL aggregation, Python trend analysis, and a live dashboard, significantly reducing turnaround time.